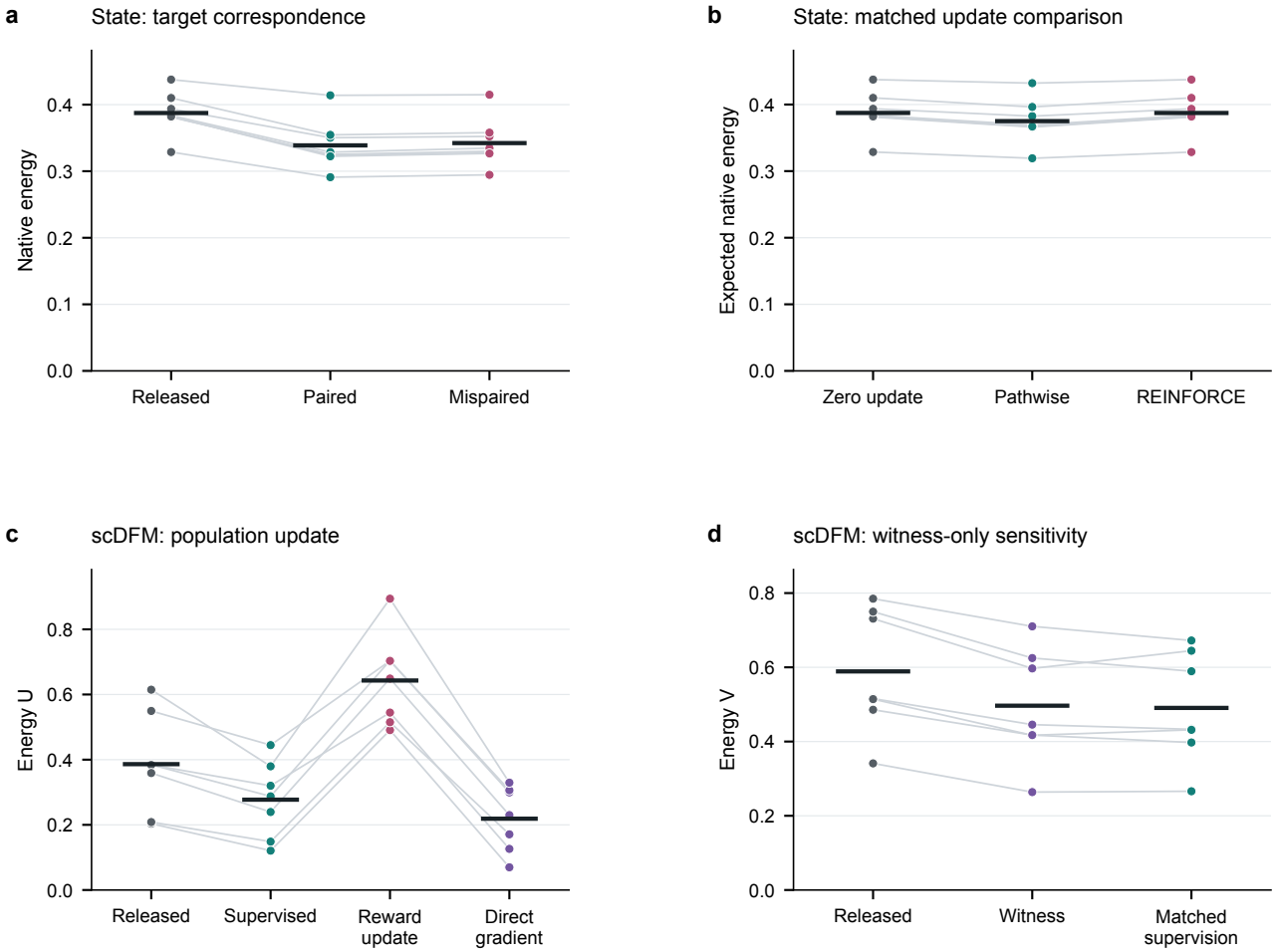
a
State: target correspondence
Native energy
0.0
0.1
0.2
0.3
0.4
Released
Paired
Mispaired
b
State: matched update comparison
Expected native energy
0.0
0.1
0.2
0.3
0.4
Zero update
Pathwise
REINFORCE
c
scDFM: population update
Energy U
0.0
0.2
0.4
0.6
0.8
Released
Supervised
Reward update
Direct gradient
d
scDFM: witness-only sensitivity
Energy V
0.0
0.2
0.4
0.6
0.8
Released
Witness
Matched supervision